

Decision Month t

One complete PPO episode

1 Observe

Fixed-size asset-feature tensor

0.42	-0.18	0.67	0.25	0.51
0.31	0.46	-0.09	0.72	0.38
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
0.58	0.12	0.41	-0.22	0.63

N active rows

Feature columns

Mask

•
•
$\vdots$
•

Marks active rows

2 Assign scores

PPO policy

Risk-score vector

sort

3 Rank assets

Predicted ranking

1	low risk
2	
$\vdots$	
$N-1$	
N	high risk

Realized-risk ranking

1	low risk
2	
$\vdots$	
$N-1$	
N	high risk

compare complete rankings

4 Evaluate ranking

One month-level reward

Evaluates the complete monthly ranking

The episode starts with one monthly universe and ends with one reward for its complete ranking.